Chi-Square Distribution

Recap

Typically, the probability distribution function for a distribution is regulated by two parameters. For the normal:

- mean μ
- standard deviation σ

These parameters are sometimes independent (normal, t) and sometimes associated with one another (binomial)

What have we been doing all year?

- Take a normally distributed variable
- Get deviation scores
- Square the deviation scores
- Sum them up

What are the consequences of doing so?

Chi-Square Distribution

$\chi^2 = \sum_{i=1}^{k} Z_i^2$

- Z_i = standard normal variable
- k = number of independent variables (**degrees of freedom**)

In English:

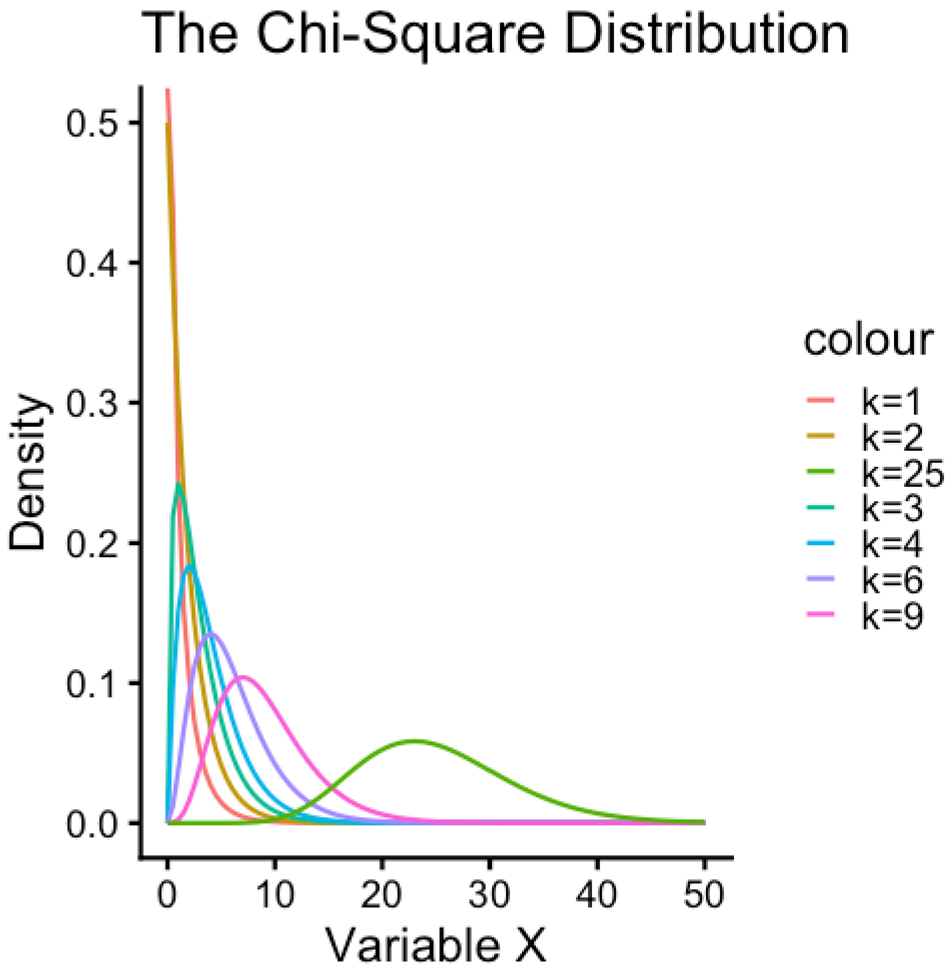
> χ^2 is created by squaring standard normal variables and adding them together

Chi-Square Distribution

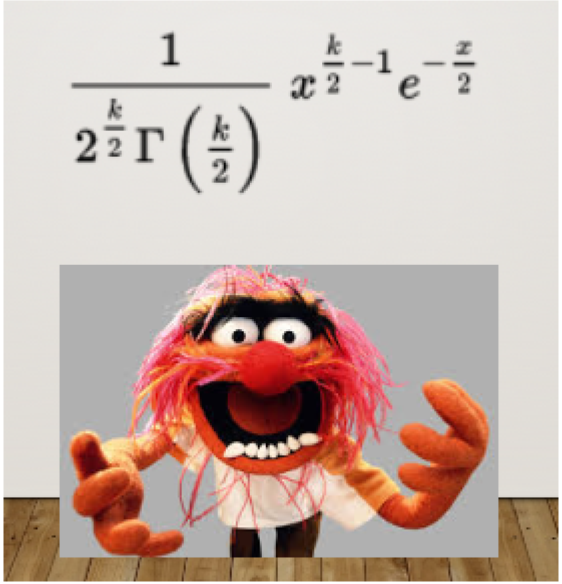
The **chi-square** (χ^2) distribution is very unique in that we really only have one parameter, k, which are your **degrees of freedom**

- mean = k
- variance = $2k$

It is entirely defined by degrees of freedom! It is the *number of pieces of information that you have*


The Chi-Square Distribution
Density
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
Variable X
colour
k=1
k=2
k=25
k=3
k=4
k=6
k=9

What kind of crazy function would lead to a distribution like this?!?!!



If a tree falls in the forest, and no one is around, does it make a sound?

Trees Example

Let's say a total of 110 trees have fallen. We might *expect* that half of those trees make a sound and half of them don't.

- 55 trees fall and do make a sound
- 55 trees fall and do not make a sound

Trees Example

We expect 55 trees to fall per category (sound and no sound). However, our data show that 10 fell and made a sound whereas 100 fell and did not make a sound.

This let's build a table that shows us what we have:

```r
trees = data.frame(TreesFallen = c(10, 100),
                   Expected = c(55, 55))
rownames(trees) = c("Registered Noise", "No Registered Noise")

trees
```

```
##                     TreesFallen Expected
## Registered Noise             10       55
## No Registered Noise         100       55
```

Trees Example

```
##                            TreesFallen Expected
## Registered Noise                    10       55
## No Registered Noise                100       55
```

$$\chi^2_P = \Sigma \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Trees Example

```
##                          TreesFallen Expected
## Registered Noise                  10       55
## No Registered Noise              100       55
```

Step 1: Get the ratio across rows:

$$\frac{(10-55)^2}{55} \quad \frac{(100-55)^2}{55}$$

```r
noise <- ((10-55)^2)/55
noise
```

```
## [1] 36.81818
```

```r
noNoise <- ((100-55)^2)/55
noNoise
```

```
## [1] 36.81818
```

Trees Example

Step 2: Sum these up

```
noise + noNoise
```

```
## [1] 73.63636
```

Trees Example

Step 3: Determine Degrees of Freedom

df = $(r-1)(c-1)$

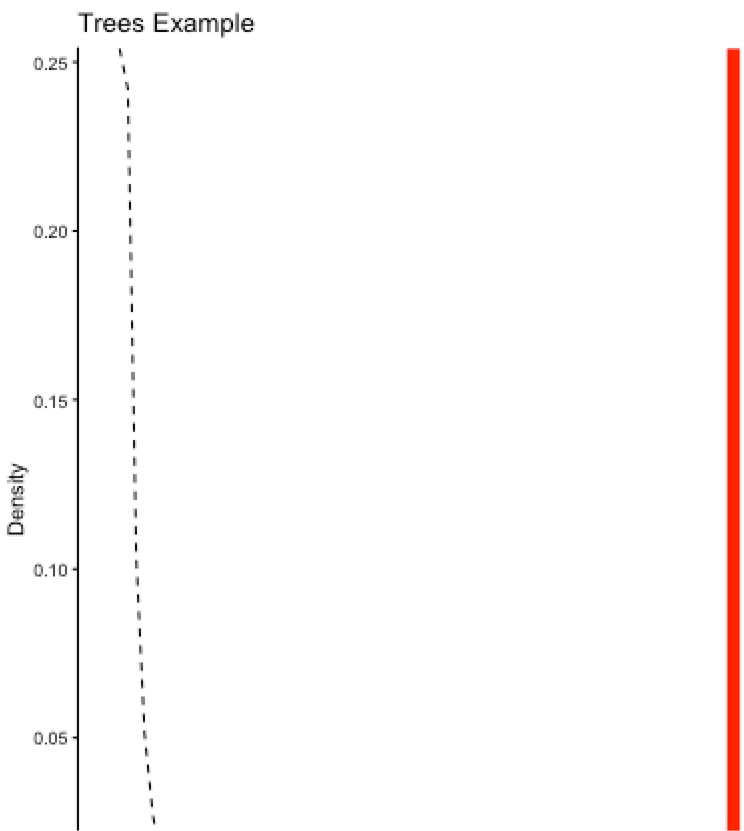
df = $(2-1)(2-1)$

df = 1

Trees Example

$\chi^2(1) = 73.636$

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warn
## generated.
```



What did we learn?

The χ^2 distribution gives us the probability to find the data (or those more extreme) given a theory (expected values)

Comparing that against some threshold for determining whether that probability is within the range of expectations or not

```
pchisq(q = 73.636, df = 1, lower.tail = FALSE)
```

```
## [1] 9.393574e-18
```

Our calculation reflects our value minus some expectation, to understand how far away it is from that expectation.

Interim Summary

- We are comparing observed counts to what we would expect from under a model:
    - If the observed = expected, χ^2 is very small
    - If observed is really different from expected, χ^2 is large
    - Larger χ^2 statistic, given some df parameter, the more likely it is that this model is wrong -- that our data come from something else
- The χ^2 test statistic is a weighted squared error statistic!

What do we want to know?

Are variables related?

- Chi-square test of association / independence (contingency tables)

Does the data match what a model predicts?

- Goodness-of-fit test

Does a more complex model improve fit?

- Likelihood ratio test

Does the model adequately explain the data?

- Deviance statistic

What do we want to know?

All of these ask the same underlying question:

> How far are the observed data from what the model predicts?

$$\chi^2_P = \Sigma \frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Let's go back...

Sometimes we want to know...are two categorical variables related to one another?

- Intervention (yes/no) & Outcome (good/bad)

If we want to to know if there's a difference

- Chi-square test of association

If we want to know the magnitude of the association?

- Odds Ratio

Effect Sizes for Chi-Square

Odds Ratio OR = number experiencing event divided by number who did not experience event.

	Outcome Yes	**Outcome No**
Exposure Yes	a	b
Exposure No	c	d

- Odds of outcome given exposure: $\frac{a}{b}$
- Odds of outcome without exposure: $\frac{c}{d}$
- **Odds Ratio**: $\frac{a/b}{c/d}$

All together

- Chi-square test is a *hypothesis* test. Gives you a p-value
- Odds ratios quantify how large the association is. It's an *effect size*
- Interpretation of OR
  - OR = 1 → no association (even odds)
  - OR > 1 → exposure increases odds of outcome
  - OR < 1 → exposure decreases odds of outcome

Cohen (1988) provided the following advice for interpreting odds ratios:

- 1.5 small
- 2.5 medium
- 4.3 large

Where are we going?

Logistic regression estimates *log odds ratios*

Next Time

- Logistic Regression
- Exam 2 on 3/24